

Jibril Afaa

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EDUCATION

The Pennsylvania State University

Bachelor of Science, Computer Science

Pittsburgh, PA

Graduation: May 2026

EXPERIENCE

Edge-Optimized Image Scaling

SME Solutions Inc

Aug 2025 – Dec 2025

State College, PA

- Engineered an end-to-end **AI super-resolution model** using **ESPCN** to upscale low-res camera streams, empirically demonstrating higher **PSNR/SSIM** than bicubic interpolation, while cutting bandwidth by **30–50%**
- Ran the model on the **Jetson Orin Nano** using an **ONNX** export produced by a teammate, enabling faster inference and stable performance on **ARM-class hardware**
- Included **Laplacian-based sharpening** to improve frame clarity on streamed video from a **Raspberry Pi**, maintaining **30FPS** after the full pipeline

Software Engineer Intern

Alli Marketplace

May 2025 – Aug 2025

Remote

- Built an orders page using TypeScript and Express, while focusing on system design
- Integrated UPS API to handle real-time shipping rates and label generation, ensuring reliable logistics processing
- Architected the backend flow to support scalable order management and smoother fulfillment

Software Engineer Intern

Axon Information Systems

May 2024 – Aug 2024

Accra, Ghana

- Architected responsive user interfaces with ReactJS and Tailwind CSS, improving usability by **20%**
- Constructed and optimized REST APIs using C# and .NET with PostgreSQL, boosting data speed by **35%**
- Collaborated with designers to deliver functional UI components **40%** faster

PROJECTS

Distributed Smart Fan Controller | STM32, Raspberry Pi Pico, FreeRTOS, C

- Built a two-node distributed embedded system with an **STM32** controller and **Raspberry Pi Pico** motor node communicating over **UART**, achieving reliable command exchange with **<5ms** round-trip latency
- Implemented a **FreeRTOS** application with dedicated tasks for input handling, state management, and inter-node communication, plus a **4-mode FSM** (Off, Low, High, Auto) bare-metal **PWM** firmware on the Pico drove DC motor speed with **±2%** duty-cycle accuracy

SSD1306 OLED Driver | STM32, I2C, C, HAL

- Wrote a complete **SSD1306 OLED driver** from scratch on the **STM32 Nucleo F411RE** by reading the datasheet directly, implementing the full **I2C initialization sequence**, command/data framing, and a **128×64 framebuffer** with manual page/column addressing
- Built a **graphics primitive layer** supporting arbitrary line drawing and real-time animations, achieving smooth rendering via full framebuffer flush over **I2C** at each frame

CS Education Content Creator | TikTok, Instagram

- Produce short-form video content on **low-level C programming** and **concurrency**, covering multithreading, race conditions, mutexes, and deadlocks, with **30K+ total views** across TikTok and Instagram
- Create walkthroughs on writing **device drivers** and embedded systems concepts, building a growing audience of CS students and early-career engineers

RELEVANT SKILLS

Programming Languages: C, C++, Python

Tools & Technologies: Git, Linux, CMake, GDB, STM32CubeIDE, OpenCV, PyTorch

Embedded Systems: STM32 (ARM Cortex-M), FreeRTOS, I2C, SPI, UART, interrupt-driven firmware